

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VI- Semester of B. Sc. (Biotechnology/Microbiology/Biochemistry)

Examination April-May 2019

Course code & Name BE 602: Pharmaceutical Quality Control and Quality Assurance

Date: 03-05-2019 Day: Friday Time: 1:30 PM To 3:30 PM Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q - II Answer the following questions as directed		
Q- II (A)	Answer <u>any TWO</u> of the following (i) Explain - (a) Disc diffusion method and Agar Cup Method (b) Big blue transgenic rodent mutation assay (ii) Explain- (a) Comet assay (b) Ames Test (iii) What is the significance of Pharmacopoeia? What are the Inclusion criteria and exclusion criteria of drugs as per Indian pharmacopoeia?	8
Q- II (B)	Answer <u>any ONE</u> of the following (i) Mention names of any four quality control tests for pharmaceutical products. (ii) Give names of antibiotics of β -lactum and aminoglycosides categories (two names for each category)	2
SECTION - II		
Q-III Answer the following questions as directed		
Q- III (A)	Answer <u>any ONE</u> of the following (i) Write down the differences between Quality Control and Quality Assurance. Explain different tests for sterility of pharmaceutical product in brief. (ii) A student wants to estimate the potency of antibiotics. Mention any three <i>in vitro</i> used to determine potency of antibiotics and what parameters under microbiological assay are used to define potency of antibiotics.	4
Q- III (B)	Answer <u>any THREE</u> of the following (i) LAL test and MAT test for pyrogens	9

	(ii) Mode of action of β-lactum antibiotics and aminoglycosides (iii) Short note on (a) Pyrogens and Endotoxins (b) Mutagens and Carcinogens (iv) Write down the importance of Cyanocobalamin in food/medicine. Explain the bioassay for Cyanocobalamin.	
Q- III (C)	Answer <u>any ONE</u> of the following (i) What are the basic duties of Quality Assurance in Pharmaceutical Industries? (ii) Mention the names of three culture media which have been designed for sterility test.	2